9 (a) (i)

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = (p(2p+16)) + ((-2)\times(-3)) + (4\times 6)$$

$$= 2p^2 + 16p + 6 + 24$$

$$= 2p^2 + 16p + 30$$

(ii)

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = 0$$

$$2p^2 + 16p + 30 = 0$$

$$2(p^2 + 8p + 15) = 0$$

$$2(p+3)(p+5) = 0$$

$p+3=0$ $\quad$ $p+5=0$

$p = -3$ $\quad$ $p = -5$

(b)

$$3\underset{\sim}{u} = 2\underset{\sim}{v}$$

$$3p = 2(2p+16)$$

$$3p = 4p + 32$$

$$-p = 32$$

$$p = -32$$

Question			Generic scheme	Illustrative scheme	Max mark
9.	(a)	(i)	•1 form an expression	•1 $p(2p+16)+(-2)(-3)+(4)(6)$	1
		(ii)	•2 equate scalar product to 0 •3 factorise •4 state values of p	•2 $p(2p+16)+(-2)(-3)+(4)(6)=0$ •3 $2(p+5)(p+3)$ •4 -5 and -3	3

Notes:

1. Evidence for •1 may appear in part (a)(ii).
2. The appearance of '$\mathbf{u}\cdot\mathbf{v}=0$' alone is insufficient for •2.
3. For •2 to be awarded '= 0' must appear at •2 or •3.
4. Do not penalise the absence of the common factor at •3.

Commonly Observed Responses:

Candidate A - incorrect expression at •2

(i) $p(2p+16)+(-2)(-3)+(4)(6)$ •1 ✓

$=2p^2+16p+30$

$=p^2+8p+15$

(ii) $p^2+8p+15=0$ •2 ✗

$(p+5)(p+3)=0$ •3 ✓ 1

$p=-5,\ p=-3$ •4 ✓ 1

Candidate B - incorrect expression at •2

(i) $p(2p+16)+(-2)(-3)+(4)(6)$ •1 ✓

$=2p^2+16p+30$

(ii) $p^2+8p+15=0$ •2 ✗

$(p+5)(p+3)=0$ •3 ✓ 1

$p=-5,\ p=-3$ •4 ✓ 1

Candidate C - incorrect expression at •2

$p(2p+16)+(-2)(-3)+(4)(6)$ •1 ✓

$2p^2+16p+24=0$ •2 ✗

$2(p+6)(p+2)$ •3 ✓ 1

$p=-6,\ p=-2$ •4 ✓ 1

Candidate D

(i) $\mathbf{u}.\mathbf{v}=\begin{pmatrix}2p^2+16p\\6\\24\end{pmatrix}$ •1 ✗

(ii) $p(2p+16)+6+24=0$ •2 ✓

$2p^2+16p+30=0$

$(p+5)(p+3)=0$ •3 ✓

$p=-5,\ p=-3$ •4 ✓

Question			Generic scheme	Illustrative scheme	Max mark
	(b)		•5 interpret relationship •6 determine value of p	•5 $3(p)=2(2p+16)$ or $3\mathbf{u}=2\mathbf{v}$ or equivalent •6 -32	2

Notes:

Commonly Observed Responses:

Candidate E

For parallel vectors $\theta = 0°$

Using $\mathbf{u}.\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$

$p(2p+16)+(-2)(-3)+(4)(6)=\sqrt{p^2+(-2)^2+4^2}\sqrt{(2p+16)^2+(-3)^2+6^2}$ •5 ✓

$p^2+64p+1024=0$

$p=-32$ •6 ✓